Executive Summary: Teleco Customer Churn Analysis

1. Project Overview

The objective of this project is to analyze customer churn in the telecommunications industry and identify the customer segments, service-related factors, and contractual patterns associated with customer attrition. The analysis was performed using Python, Pandas, Matplotlib, and Seaborn, with visualizations designed to transform raw customer data into actionable business insights.

The dataset contains *7,043 customer records and 21 columns*, covering customer demographics, tenure, phone and internet services, additional service subscriptions, contract types, payment methods, and billing information. The exploratory data analysis (EDA) included data inspection, missing-value checks, duplicate identification, data-type conversion, and visualization of churn patterns across multiple customer attributes.

The analysis focuses on understanding the overall churn rate, comparing customer segments, identifying potential risk factors, and recommending strategies that could help the company improve retention and customer lifetime value.

2. Overall Customer Churn

The overall churn distribution reveals that approximately 26.54% of customers have discontinued their services , while 73.46% remain with the company.

- Total customers analyzed: 7,043
- Customers who churned: 1,869 (26.54%)
- Customers who stayed: 5,174 (73.46%)

This means that approximately **one out of every four customers** has left the company. Although most customers remain active, the proportion of customers leaving represents a significant customer-retention challenge.

The countplot clearly shows the difference between retained and churned customers, while the pie chart provides a percentage-based view of the overall churn distribution. Together, these visualizations establish the baseline against which individual customer segments can be evaluated.

From a business perspective, reducing churn even modestly could help preserve recurring revenue, reduce customer-acquisition costs, and improve the long-term value generated by existing customers.

3. Customer Demographics and Churn

Gender-based analysis

The gender-based countplot compares male and female customers across churn categories. The distribution suggests that churn occurs among both genders, without a major difference apparent from the overall counts.

This indicates that gender alone may not be a particularly useful variable for prioritizing retention campaigns. However, a definitive conclusion requires comparing churn rates within each gender rather than relying only on customer counts.

Business implication: The company should prioritize more actionable characteristics, such as contract type, tenure, payment behavior, and service experience, rather than using gender as a primary retention criterion.

Senior citizens

The percentage-stacked bar chart comparing senior and non-senior customers provides a more meaningful comparison because each bar represents 100% of its respective customer segment.

The chart indicates that the proportion of churned customers is higher among senior citizens than among non-senior customers. In the standard Telco Customer Churn dataset, the approximate churn rates are:

- Senior citizens: 41.7%
- Non-senior customers: 23.6%

The difference is approximately 18.1 percentage points, suggesting that senior customers may represent a higher-risk segment.

Business implication:The company could investigate whether senior customers experience difficulties with service setup, billing, technical support, or understanding their plans. Personalized assistance and accessible customer support may help improve their experience.

These differences should be interpreted as associations, not proof that age itself causes customers to churn.

4. Customer Tenure and Churn

The tenure histogram separates customers by the number of months they have used the company's services and distinguishes between retained and churned customers.

Your visualization shows a clear pattern: churn is concentrated among customers with shorter tenure, particularly those who have used the service for only one or two months. In contrast, customers with longer service histories are more likely to remain with the company.

This suggests that the early stages of the customer lifecycle are particularly important for retention. New customers may still be evaluating service quality, pricing, network performance, and whether the product meets their expectations.

As customers remain with the company, they may become more familiar with its services and develop greater satisfaction or switching costs.

Business implication: The company should introduce an early-life retention strategy that includes:

- Welcome messages and onboarding assistance during the first month.
- Follow-up communication after installation or service activation.
- Early identification of complaints, service interruptions, or billing confusion.
- Targeted offers and support for customers showing signs of dissatisfaction.

Monitoring churn within specific tenure bands, such as 0–3 months, 4–12 months, and more than 12 months, would make this analysis even more actionable. Exact churn percentages for these bands should be calculated from the dataset rather than inferred from the histogram alone.

5. Contract Type and Churn

The contract-based countplot compares churned and retained customers across month-to-month, one-year, and two-year contracts.

The visualization indicates that month-to-month customers account for the largest number of churned customers. Customers on longer-term contracts are more likely to remain with the company.

Approximate churn rates in the standard dataset are:

Contract type	*Approximate churn rate*
Month-to-month	42.7%
One year	11.3%
Two years	2.8%

These rates suggest that month-to-month customers have substantially higher churn than customers with longer contractual commitments.

The difference may reflect greater flexibility to switch providers, differences in customer loyalty, or variations in the types of customers choosing each contract. The chart alone cannot establish which explanation is responsible.

Business implication: The company could encourage customers to consider longer-term plans through transparent pricing, loyalty benefits, service bundles, and discounts that provide genuine value. Before introducing incentives, the company should assess their effect on profit margins and customer satisfaction.

Retention efforts should also address why month-to-month customers leave, rather than assuming that contractual commitment alone will solve the problem.

6. Phone and Internet Services

Your nine-panel countplot analysis examines PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies.

This is a useful part of the project because it compares churn across several service-related variables in a single visual layout.

PhoneService and MultipleLines

PhoneService has a large customer base, so its churn counts should be interpreted in the context of its segment size. A high count of churned customers does not necessarily mean that a service has a high churn rate.

Similarly, MultipleLines should be evaluated by comparing the proportion of customers who churn within each category, rather than comparing the raw number of departures alone.

InternetService

Fiber Optic customers account for a substantial number of churned customers. In the standard dataset, the approximate churn rates are:

Fiber Optic	41.9%
DSL	19.0%
No internet service	7.4%

This pattern makes Fiber Optic customers an important segment for further investigation. Possible areas to examine include pricing, perceived service quality, reliability, customer expectations, and the value offered by competing providers.

The results do not establish that Fiber Optic technology itself causes churn. Differences in pricing, customer profiles, and subscribed services may also contribute.

OnlineSecurity and TechSupport

The charts indicate noticeably higher churn counts among customers who do not subscribe to Online Security or Tech Support.

Approximate churn rates in the standard dataset are:

Service status	*Churn rate*
No Online Security	41.8%
Has Online Security	14.6%
No Tech Support	41.6%
Has Tech Support	15.2%

Customers without these services show substantially higher churn rates. This makes both variables important candidates for further retention analysis.

However, customers who purchase additional services may differ in tenure, engagement, or overall satisfaction. Therefore, these relationships should not automatically be interpreted as evidence that purchasing the services directly prevents churn.

Business implication: The company could review the needs of customers without Online Security or Tech Support, communicate the benefits of relevant service bundles, and improve support accessibility. Fiber Optic customers should also be examined for recurring complaints, pricing concerns, and service-quality issues.

OnlineBackup, DeviceProtection, StreamingTV, and StreamingMovies

These variables provide additional insight into customers' service preferences and subscription patterns. The countplots can help identify which categories contain large numbers of churned customers.

For a stronger conclusion, each service should be assessed using category-level churn rates. This would distinguish between a service category that has many churned customers because it is popular and one that has a genuinely elevated risk of churn.

7. Payment Method and Churn

The payment-method countplot compares churned and retained customers across electronic checks, mailed checks, bank transfers, and credit-card payments.

The visualization highlights electronic-check users as an important customer segment because they account for a high number of churned customers.

In the standard dataset, approximate churn rates are:

Payment method	*Approximate churn rate*
Electronic check	45.3%
Mailed check	19.1%
Bank transfer (automatic)	16.7%
Credit card (automatic)	15.2%

Electronic-check users exhibit a considerably higher churn rate than customers using the other payment methods. This makes payment behavior a valuable variable for identifying potentially at-risk customers.

Nevertheless, the analysis does not establish that electronic checks cause customers to leave. Payment preferences may be associated with other characteristics, including contract type, tenure, or customer demographics.

Business implication: The company should investigate the electronic-check segment to understand whether billing confusion, payment friction, or other customer-experience issues contribute to churn. It could also encourage convenient automatic payment options where appropriate, while ensuring that customers retain a fair choice of payment methods.

8. Data Quality and Preparation

Before performing the visual analysis, the notebook includes several useful data-preparation steps:

- Inspected the dataset structure using `df.info()` and summarized numerical variables using `df.describe()`.
- Checked for missing values and found zero missing values after the data-cleaning steps.
- Checked customer identifiers and found no duplicate customer IDs.
- Converted `TotalCharges` from an object data type to a numeric data type.
- Replaced blank `TotalCharges` entries with zero, based on the assumption that those customers had zero tenure and no recorded charges.
- Converted the `SeniorCitizen` indicator from numeric values to readable Yes/No categories.

These steps improved the readability and usability of the dataset for visualization.

One consideration is that the blank `TotalCharges` values should ideally be validated against the corresponding tenure and billing records before being replaced. This ensures that the cleaning decision is supported by the underlying data rather than an assumption alone.

9. Key Business Recommendations

Based on the observed patterns, the following actions could help the company develop a more effective customer-retention strategy.

1. Prioritize early-stage customers - Introduce onboarding support, proactive communication, and early issue resolution to address the higher churn observed among customers with short tenure.

2. Improve month-to-month customer retention - Investigate why flexible-contract customers leave and test attractive longer-term plans or loyalty benefits.

3. Investigate Fiber Optic customer experience - Review pricing, reliability, complaints, and service expectations to understand the elevated churn rate in this segment.

4. Review electronic-check users - Examine billing experiences and payment preferences to determine whether payment-related friction is associated with customer departures.

5. Improve the value of support and security services - Make Online Security and Tech Support benefits easier to understand and assess whether suitable bundles improve customer satisfaction and retention.

6. Develop targeted support for senior customers - Investigate potential barriers to service use and provide accessible support based on actual customer needs.

7. Monitor churn using percentage-based KPIs - Track overall churn rate, tenure-based churn, contract-based churn, payment-method churn, and service-specific churn over time. This will make it easier to determine whether retention initiatives are working.

10. Conclusion

This exploratory analysis identifies several important patterns in telecom customer churn. Approximately 26.54% of customers have left the company, with higher churn associated with short tenure, month-to-month contracts, senior-citizen status, Fiber Optic internet service, the absence of Online Security and Tech Support, and electronic-check payments.

The findings suggest that customer retention should focus on the early customer lifecycle, the experience of flexible-contract customers, and service and billing segments that exhibit elevated churn. The combination of data cleaning, countplots, a percentage pie chart, a percentage-stacked bar chart, a tenure histogram, and multi-panel service visualizations provides a useful foundation for understanding these patterns.

The next step is to validate segment-level churn rates directly against the dataset, investigate interactions between customer attributes, and develop a predictive model to identify customers at risk of leaving. By combining descriptive analysis with targeted experiments and ongoing monitoring, the company can move toward more effective, measurable, and data-driven customer-retention strategies.